

Answer Key — Storms, Lightning and Cyclones

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) thunderstorm** — a thunderstorm has lightning and thunder.
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- Q2 **(b) eye** — the eye of the cyclone is calm and has the lowest pressure.
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- Q3 **(b) warm ocean waters** — warm ocean water supplies the moist rising air.
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- Q4 **(c) crouch low in an open area** — crouch low, away from tall objects; don't lie flat.
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- Q5 **(a) Both true, R explains A** — the rapid heating and expansion of air is exactly what makes thunder.
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- Q6 **conductor** — a lightning conductor protects buildings.
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- Q7 **rotation** — Earth's rotation makes the moving air spin.
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Short answer

- Q8 Strong up/down winds rub water droplets and ice particles, building static charges; charges separate, and when very large the air's insulation breaks down, giving a sudden flow of charge — a flash of lightning.
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- Q9 A large spinning system of clouds, wind and rain around a very low-pressure centre; it forms over warm ocean waters.
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- Q10 Stay away from tall objects, crouch in a low open area (don't lie flat), avoid metal-rod umbrellas, get out of water, shelter in a car/bus (any three).
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Long / application

- Q11 Warm moist air rises from the warm ocean; water vapour condenses releasing heat, warming the air more so it rises further, forming a very low-pressure centre. Surrounding air rushes in and Earth's rotation makes it spin. The eye is the lowest-pressure centre where wind is calm (strong winds revolve around it). Over land the warm-moist-air supply is cut off, so the cyclone weakens.
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- Q12 Warm moist air rises and condenses into towering clouds; up/down winds rub droplets and ice, separating charges. A huge charge breaks down air's insulation, giving lightning; the lightning heats air which expands rapidly, making thunder. A lightning conductor gives the charge an easy metal path to the ground, protecting the building.
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